Where groups disagree

Leases (C6, B1): M4.3 uses 8× rent; F5 and the PDF use the filed liability; S&P uses the present value at 10%; Kricheff counts only capital leases. **Wins:** the filed liability, because after ASC 842 it is already a present value of the remaining term; S&P's 10% present value when no liability is tagged; 8× for old periods only. The code already uses the liability. Pensions (C7, B2): F5 adds the deficit to debt; the PDF also adds expense less service cost to EBITDA; S&P adds PBO plus OPEB less plan assets, after tax. **Wins:** the PDF form with OPEB included; apply the tax shield only for a cash tax payer, because most HY issuers pay little cash tax (Kricheff). FCCR (K4): **wins** F5 for lease-heavy issuers, the skill form for amortising loans; print both. Headroom bands (K5, B6): **wins** F8 bands on the F8 formula, with the G3 trajectory rule and Kricheff's one-year-out check. Creation leverage (C5): **wins** the PDF form.

Settling test

MGEC through-unsecured creation leverage must equal 4.27x (PDF U7). Lease gap by archetype: where the methods differ by more than 0.5x, keep the one that tracks the spread residual better out of sample. Breach rate by headroom band on matured loan tests.

Status

Partial. Built: gross, net and lease-adjusted leverage (liability method), coverage, FCCR without amortisation, bond table. Missing: face by tranche and rank, creation leverage, FFO and FOCF ratios, pension and OPEB, notching, all covenant work, the claims view, the LME score. Check that the debt tag excludes finance leases, or they count twice.

CRWV

Debt 35,551m; lease liabilities 16,540m; leverage 9.45x; lease-adjusted 8.94x; net 7.99x; after-capex coverage −12.79 (capex exceeds EBITDA); FCCR 0.54. Four senior unsecured bonds, 2030 to 2032, at 88.8 to 93.7 and 661 to 776 bp. 3.7bn of debt sits in ring-fenced vehicles that hold 18.2bn of assets: structurally ahead of the parent bonds on those assets.

6 S6 Two clocks

Purpose. Compare when fixed charges fall due with when signed contracts pay, to size the part of the debt that is a bet on renewal.

What each group contributes

- **G1** M4.7 gives the formulas, the features and five rules (read schedules as filed; a 10-Q ladder starts with the rest of the fiscal year; state every assumption; build the downside from the main risk flag; read the shortfall as a lower bound). Gatto has no such measure; the nearest are the maturity schedule (Figure 8.4, p. 229) and leases (pp. 267–268). The PDF has none either; its closest rule is R53 (forecast to the next material contractual change).
- **G2** Built: `hy/clocks.py`, `scripts/clock_case.py`, the prose parsers in `hy/sources/clock_inputs.py`. Source card: Radhakrishnan (2026): AI capacity providers carry long fixed leases that start on dates they do not control, against shorter and concentrated contracts; unsecured debt is a bet on renewal.
- **G3** `credit-model-standards`: for customer concentration, model the loss of the top customer (revenue, margin, working capital). The clocks run this as their downside.
- **G4** S&P (Ch. 6, 7): forecast beyond the maturity when refinancing risk exists; acceleration and cure timing (30 days for interest). Kricheff (Ch. 8, 13, 23): accreting and PIK debt raise the claim without cash; bond coupons are lumpy (semiannual), so map cash against payment dates in distress.

Algorithm

1. Put every fixed obligation on one yearly grid from the filing date: leases in force, signed leases not started, debt principal, interest on the falling balance, purchase commitments. Read each ladder as filed; check it against its stated total.
2. Put the RPO on the same grid by the filer's stated windows; spread the rest over a stated tail.
3. Choose the margin m as the EBITDAR margin; add a 20% stress margin.
4. Compute the centres D_O, D_R, the gap, the breakeven margin, cover, shortfall, the renewal-dependent share and its size in years of EBITDA.
5. Downside: remove the top customer's share of RPO (customer shares from XBRL; RPO split assumed equal to revenue split).
6. If RPO is below one year of revenue, skip the revenue clock and report the cost clock against today's EBITDAR held flat.
7. Pass `revenue_runoff` and the largest contract expiry to S12 as event dates.

Metrics

$$D_O = \frac{\sum_t (t+\frac{1}{2})O_t}{\sum_t O_t}, \quad D_R = \frac{\sum_t (t+\frac{1}{2})R_t}{\sum_t R_t}, \quad gap = D_O - D_R, \quad m^* = \frac{\sum_t O_t}{\sum_t R_t}. \tag{17}$$

Intuition. the centres of the two clocks; how many years the charges outlast the contracts; and the margin at which signed contracts pay every charge (above 1, no margin is enough).

Symbols. O_t fixed obligations in year t; R_t RPO booked in year t.

$$\begin{aligned} cover = \frac{m\sum_t R_t}{\sum_t O_t}, \qquad short = \max\Big(0, -\min_t \sum_{s\le t}(mR_s - O_s)\Big), \\ share = \frac{\sum_{t>T^{run}} O_t}{\sum_t O_t}, \qquad yrs = \frac{share \cdot \sum_t O_t}{E}. \end{aligned} \tag{18}$$

Intuition. the share of charges that contracts pay; the largest cash hole that renewals or new debt must fill; the share due after the contract book is gone; and that amount in years of today's EBITDA.

Symbols. m EBITDAR margin; T^{run} year by which 99% of RPO is booked; E LTM EBITDA.

Bands

m^*: < 0.6 / 0.6–1.0 / > 1.0. *cover*: > 1.0 / 0.7–1.0 / < 0.7. *share*: $< 15\%$ / 15–30% / $> 30\%$. *yrs*: < 2 / 2–5 / > 5. All v0.

Inputs

Data: XBRL ladders, uncommenced lease total, RPO, customer shares, LTM EBITDA and lease cost. *Documents:* 10-Q prose for commencement windows, lease terms, RPO windows. *Analyst:* tail lengths, RPO timing after the last window, the stress margin.

Outputs

`duration_gap`, `breakeven_margin`, `coverage_contracted`, `renewal_dependent_share`, `renewal_dependent_ebitda_years`, `max_cum_shortfall`, `years_of_ebitdar`, scenario and sensitivity tables.

Where groups disagree

No group disagrees on the measure. One gap: M4.7 says three clock features enter the fair spread model. The fair line uses only leverage. **Rule:** until the M7.2 model exists, the clocks enter as estimate E5 of the combined score and as risk metrics in the pair (S9).

Settling test

Decision page `2026-10-22-duration-mismatch`: `renewal_dependent_ebitda_years` must have a positive partial correlation with spread after the M7.2 features, and a positive rank IC for spread change around renewal dates. Code tests: `tests/test_clocks.py` (11) and `tests/test_clock_inputs.py` (19).

Status

Built. Missing only the link into a fair spread model.

CRWV

Charges 106.4bn against RPO 103.7bn; centres at year 5.5 and 2.6 (gap 2.9 years); breakeven margin 1.03 against an EBITDAR margin of 0.77; cover 0.75; 31% of charges (32.6bn) after the runoff; largest shortfall 26.7bn. If the top customer stops paying: cover 0.48, shortfall 55.4bn.

7 S7 Economic stories and cases

Purpose. Project the issuer under several company cases inside five macro stories, and weight them so the expected numbers carry the odds of each story.

What each group contributes

- **G1** Gatto step 4 (Chapter 10, pp. 283–318): management, base and downside cases (p. 285); the base case is a coin toss (p. 285); probabilities by judgement (p. 311); if the downside looks likely, it becomes the base (p. 311). M5.1 engine; M5.2 macro logistic of a large EBITDA fall; M5.3 forward features; M5.4 forecast linkage (registry, five paths, Bayes update, refinance at the forward Treasury plus fair spread). F6: six cases, mechanics F6.1–F6.5, phase vote and weight table, archetype multipliers, F6.6 promotion, F6.9 fallback order, F6.10 Bayes. PDF: horizon must reach the next material refinancing (R53); eight forecast items (P10); separate cash and face-debt recursions (E38, E39); states must include default (R60); reverse stress.
- **G2** Built: the forecast registry (`hy/forecasts.py`); the path builder (`hy/macro.py`); the case grid (`hy/forward.py`); `scripts/scenarios.py` with its v0 defaults. Forecast cards: SPF medians and density bins, FOMC SEP, Treasury forwards, SLOOS, TRACE HY level are live; agency default forecasts, fed funds futures, ICE OAS, guidance, consensus and sector decks are stubs.
- **G3** `credit-model-standards` and `hy-model-standards`: the downside comes from the issuer's own main risk; a severe case combines a revenue fall, a margin squeeze and working-capital stress; report first breach quarter, peak leverage, trough FCF and runway; tag a recovery shape (`v`, `u`, structural, customer replacement); roll debt per tranche; interest on last quarter's debt, no solver. Adaptation: the "lender case with a conservative tilt" does not transfer (K3); public HY guidance leans optimistic.
- **G4** S&P (Ch. 6, 9): base, stress and default cases; stress the worst credible case, not a soft one ("it is too easy to remain too soft"); front-load a 10–30% revenue drop with fixed costs for high-risk issuers; cyclicals use their historical band (Honda: EBITDA down about 50%; volumes −20%, −10%, +5%, +10%, +15%); the default point is $E - T^c - K < I + Am$; three default types set how far cash flow falls. Kricheff (Ch. 12, 13, 21): drive revenue top down; run solve-for cases (the EBITDA fall that makes FCF negative); a rate shock for floating debt and refinancing; event horizons of 6–18 months with probabilities; base model pays scheduled amortisation and keeps the rest as cash.

Algorithm

1. Read the registry at t. For GDP and CPI, fit a normal per horizon from the 10th and 90th percentiles, widened by each source's error record.